

Arnav Tripathi

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SUMMARY

Computer Science undergraduate pursuing backend engineering internships, with hands-on experience building FastAPI services, real-time webhook pipelines, and OAuth integrations in a live production environment. Additional project experience spanning generative AI applications, computer vision, and machine learning.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

Frameworks: FastAPI, React 19, SQLAlchemy, TensorFlow/Keras, APScheduler

Backend & Systems: PostgreSQL, WebSockets, REST APIs, Webhooks, , Neo4j

AI/ML: Random Forest, Generative AI

Tools: VS Code, Git/GitHub, Figma, Tableau — **Cloud:** AWS

Data & Analytics: Data Analysis, Prompt Engineering, Statistical Modelling

PROJECTS

Smart Traffic Violation Reporting System [Github](#)

2024

- Secured Second Runner-up at SOLVIT Hackathon competing against 150+ teams
- Built a system using computer vision to detect violations (speeding, red-light running, no seat belt) in real time with 90% detection accuracy
- Implemented automated evidence capture (images/video) with instant alert dispatch to law enforcement agencies

Pathwise [Github](#)

Nov 2025 – Mar 2026

- Built an interactive React 19 + Vite UI with Framer Motion animations and Recharts for data visualization
- Integrated AI-generated course content with Markdown/LaTeX rendering, quizzes, and gamification features
- Implemented PDF/ZIP export functionality and OpenRouter AI integration, reducing content generation time by 85%

Network Anomaly Detection for IoT Intrusion Detection [Github](#)

2026

- Built an unsupervised intrusion detection pipeline (Isolation Forest, One-Class SVM) to flag malicious IoT traffic without labeled data, benchmarked on the BoT-IoT and NSL-KDD datasets (3M+ records)
- Engineered an SQLite analytics layer and Tableau dashboard to visualize model accuracy, precision, and recall

EXPERIENCE

Software Development Intern – Backend Engineering

May 2026 – Present

SwarnShrresti Technologies India Pvt. Ltd.

Remote

- Engineered a FastAPI backend for Orchestra, a multi-platform developer activity aggregator, supporting concurrent users via serverless PostgreSQL and SQLAlchemy ORM
- Built real-time webhook ingestion pipelines (GitHub, Discord, Figma) with HMAC-SHA256 verification and an async WebSocket-driven state machine, cutting live task-update latency to 50 ms
- Collaborated with a 5-member distributed team to resolve schema mismatches and maintain zero-downtime integration

Virtual Intern – Google Cloud Generative AI

Jul 2025 – Sep 2025

NASSCOM

Remote

- Applied LLM, GAN, and multimodal AI concepts to build guided AI solutions under mentorship from Google Cloud experts
- Applied prompt engineering and statistical modelling to design cloud-based AI deployment pipelines

EDUCATION

VIT Bhopal University

Bhopal, India

Bachelor's Degree, Computer Science – CGPA: 8.23

2023 – 2027

Macrovision Academy

Burhanpur, India

CBSE 12th Board – 74.2%

2023

Bhandari Public School

Khandwa, India

CBSE 10th Board – 92%

2021

CERTIFICATIONS

- NPTEL – Introduction to Machine Learning – IIT Madras
- NPTEL – Marketing Analytics – IIT Kharagpur
- Google – Technical IT Support Certificate
- Oracle – Certified Generative AI Professional
- Coursera – Bits and Bytes of Computer Networking